ENTERPRISE PROJECT CONTROLS & PMO STANDARD

Project Management with Earned Value Management (EVM) Handbook

Comprehensive Reference Guide: Integrating Scope, Schedule, and Cost Performance Metrics | PMI-PMBOK Standards, EIA-748 Compliance, and Modern Power BI Analytics

1. Executive Summary & Foundational Principles

Earned Value Management (EVM) is an industry-standard project management methodology that integrates scope, schedule, and cost measurements to assess project performance and predict future outcomes. Traditional financial tracking only compares *actual costs* against *planned budgets*, completely ignoring whether the work was actually performed. EVM bridges this gap by introducing the concept of "Earned Value"—measuring the physical work completed against the baseline plan.

> **Core Tenet of EVM:** You cannot effectively manage a project if you only know how much money you spent. You must know what physical progress and value were delivered for that expenditure.

2. Core EVM Variables & Definitions

Every EVM system rests upon four foundational variables established during the planning phase and updated periodically during execution:

- BAC (**Budget at Completion**): The total authorized budget for accomplishing the project or work package.
- PV (**Planned Value** / **BCWS**): The authorized budget assigned to scheduled work to be completed by a specific date ($\text{BCWS} = \text{Budgeted Cost of Work Scheduled}$).
- EV (**Earned Value** / **BCWP**): The measure of work performed expressed in terms of the budget authorized for that work ($\text{BCWP} = \text{Budgeted Cost of Work Performed}$).
- AC (**Actual Cost** / **ACWP**): The total actual costs incurred in accomplishing the work performed within a given time period ($\text{ACWP} = \text{Actual Cost of Work Performed}$).

3. Variance Analysis: Cost & Schedule

Variances quantify the absolute magnitude of deviation from the performance measurement baseline in monetary terms ($\$$).

Schedule Variance (SV)
$\text{SV} = \text{EV} - \text{PV}$

Interpretation:
- $\text{SV} > 0$: Ahead of schedule.
- $\text{SV} = 0$: Exactly on schedule.
- $\text{SV} < 0$: Behind schedule.

Cost Variance (CV)
$\text{CV} = \text{EV} - \text{AC}$

Interpretation:
- $\text{CV} > 0$: Under budget (favorable).
- $\text{CV} = 0$: Exactly on budget.
- $\text{CV} < 0$: Over budget (unfavorable).

4. Performance Efficiency Indices

While variances tell you the *amount* of deviation, indices measure the *efficiency* or rate at which the project is performing.

Index Name	Formula	Favorable Threshold	Critical Threshold & Action
Cost Performance Index (CPI)	$\text{CPI} = \frac{\text{EV}}{\text{AC}}$	$\text{CPI} \ge 1.0$	$\text{CPI} < 0.90$: Severe cost overrun. Audit labor efficiency and rework.
Schedule Performance Index (SPI)	$\text{SPI} = \frac{\text{EV}}{\text{PV}}$	$\text{SPI} \ge 1.0$	$\text{SPI} < 0.90$: Critical path slippage. Resequence activities.
Critical Ratio (CR)	$\text{CR} = \text{CPI} \times \text{SPI}$	$\text{CR} \ge 1.0$	$\text{CR} < 0.80$: PMO escalation required for corrective action.

5. Forecasting Final Cost and Time (EAC & ETC)

Forecasting uses current performance trends to predict future project outcomes. The Estimate at Completion (EAC) projects the total cost at project completion.

1. Typical Variance Performance (Most Common):
Assumes remaining work will be performed at the historical efficiency rate (CPI).
$\text{EAC} = \frac{\text{BAC}}{\text{CPI}}$

2. Atypical Variance Performance:
Assumes past variances were anomalies and future work will go according to baseline budget.
$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$

3. Combined Cost and Schedule Efficiency:
Accounts for both cost (CPI) and schedule (SPI) degradation.
$\text{EAC} = \text{AC} + \frac{\text{BAC} - \text{EV}}{\text{CPI} \times \text{SPI}}$

Estimate to Complete (ETC): The expected cost needed to finish all remaining project work.
$\text{ETC} = \text{EAC} - \text{AC}$ (or recalculated bottom-up).

Variance at Completion (VAC): The projected budget surplus or deficit at project end.
$\text{VAC} = \text{BAC} - \text{EAC}$

6. To-Complete Performance Index ($TCPI$)

The $TCPI$ measures the cost efficiency required for all remaining work to finish within a specific financial target (either the original BAC or a revised EAC).

Targeting Original Budget (BAC):
$\text{TCPI}_{\text{BAC}} = \frac{\text{BAC} - \text{EV}}{\text{BAC} - \text{AC}}$
Indicates efficiency needed to hit initial baseline.

Targeting Revised Forecast (EAC):
$\text{TCPI}_{\text{EAC}} = \frac{\text{BAC} - \text{EV}}{\text{EAC} - \text{AC}}$
Indicates efficiency needed to hit current EAC.

7. Enterprise Implementation & Modern Analytics (Excel & Power BI)

Modern project controls move away from manual spreadsheets toward integrated automated pipelines.

- **Data Architecture:** Extracting scheduling data from Primavera P6 ($.XER$ files) or Microsoft Project and actual financial ledgers from ERP systems (SAP/Oracle).
- **The S-Curve Visual:** Plotting cumulative PV, EV, and AC over project timeline to visualize variance trends instantly.
- **Power BI & DAX Modeling:** Using relational star schemas and DAX measures to compute running totals and dynamic slicers by Work Breakdown Structure (WBS) and control accounts.